

Helmet Safety Detection System Using YOLOv8

AIRI Team PITB — AI Internship Task 1
End-to-End Computer Vision Object Detection Project

Model: YOLOv8n (Ultralytics)
Platform: Google Colab + Google Drive
Classes: helmet, vest, without_helmet, without_vest
Training: 40 epochs | 640 × 640

Final Project Report

1. Problem Definition

This project develops a Computer Vision system for automated safety monitoring. The objective is to detect safety helmets and vests and identify visible safety-violation cases in construction-site and workplace images. Automated detection can support basic safety screening and reduce the effort required for continuous manual monitoring.

Target Classes

Class ID	Class	Description
0	helmet	Safety helmet
1	vest	Safety vest
2	without_helmet	Person without required helmet
3	without_vest	Person without required vest

Potential Applications

- Construction-site safety monitoring
- Industrial workplace inspection
- CCTV-based safety screening
- Automated PPE compliance support

2. Data Collection and Preparation

A suitable safety-object dataset was selected as the starting point because it already contained relevant object annotations and the required four safety classes. To make the dataset suitable for the internship project, the selected data was reviewed, filtered and reorganized into a compact final dataset.

Additional Dataset Preparation

The project was not limited to simply using the original dataset as-is. Additional suitable images were incorporated and checked so that the final dataset better represented the intended project classes and visual conditions. The added/selected images were annotated in the same YOLO format and reviewed for consistency before training.

During preparation, images were checked for duplicates, unusable samples, missing labels, inconsistent class IDs and unsuitable annotations. The final working dataset contains **200 images** and four classes.

Dataset Quality Goals

- Keep image and label pairs synchronized.
- Use consistent class IDs and class names.
- Keep bounding boxes tight around visible objects.
- Avoid duplicate or highly similar images.
- Include different viewpoints, backgrounds and object sizes.
- Review difficult images before training.

3. Annotation

The starting dataset already contained annotations for the relevant safety objects. As part of project preparation, additional selected images were annotated and brought into the same YOLO annotation format. Existing annotations were also reviewed so that the complete final dataset followed one consistent labeling scheme.

YOLO Annotation Format

`class_id x_center y_center width height`

The four class IDs were kept consistent throughout the dataset. Bounding-box coordinates are normalized between 0 and 1. Annotation review focused on tight boxes, correct class assignment, complete labeling of visible target objects, and avoiding unnecessary background inside boxes.

Final Annotation Mapping

ID	Class Name	Annotation Meaning
0	helmet	Helmet bounding box
1	vest	Vest bounding box
2	without_helmet	Without-helmet case
3	without_vest	Without-vest case

This preparation step made the final dataset suitable for YOLOv8 training and later manual error analysis.

4. Dataset Organization

The final dataset is organized into separate image and label directories for training, validation and testing.

```
dataset/  
  ■■■ images/  
    ■ ■■■ train/  
    ■ ■■■ val/  
    ■ ■■■ test/  
  ■■■ labels/  
    ■ ■■■ train/  
    ■ ■■■ val/  
    ■ ■■■ test/  
  ■■■ data.yaml
```

Dataset Item	Final Value
Total images	200
Classes	4
Image/label pairing	Maintained
Train / validation / test	Separated
Configuration	data.yaml

The project follows the internship requirement for a small, manageable Computer Vision dataset while keeping the train, validation and test data separated.

5. Dataset Configuration

The data.yaml file defines the dataset paths and the four class names used during training.

```
path: dataset
train: images/train
val: images/val
test: images/test
```

```
names:
0: helmet
1: vest
2: without_helmet
3: without_vest
```

This configuration allows Ultralytics YOLOv8 to locate the training, validation and test images and map numeric class IDs to readable class names.

6. Google Colab Environment

Training was performed in Google Colab, while Google Drive was used to keep the project files persistent. The notebook contains environment setup, Drive mounting, package installation and GPU availability checks.

Core workflow

```
from google.colab import drive
drive.mount('/content/drive')
```

```
from ultralytics import YOLO
model = YOLO('yolov8n.pt')
```

The training notebook was saved inside the project's notebooks/ directory so that the complete workflow can be reproduced from the project package.

7. Model Training

A pretrained YOLOv8n detector was fine-tuned on the prepared safety dataset. Training was performed for 40 epochs at 640 × 640 image size.

Parameter	Final Configuration
Model	YOLOv8n
Initial weights	yolov8n.pt
Epochs	40
Image size	640 × 640
Platform	Google Colab
Output	yolov8n_baseline
Final checkpoint	best.pt

Training artifacts were preserved, including results.csv, training/validation visualizations, precision/recall/F1 curves, PR curve, confusion matrices, results.png, and best.pt/last.pt.

8. Model Evaluation

The final trained model was evaluated using standard object-detection metrics.

Metric	Result
Precision	0.9517
Recall	0.9016
mAP@0.5	0.9503
mAP@0.5:0.95	0.667

Interpretation

- Precision of 0.9517 indicates that most predicted detections were correct.
- Recall of 0.9016 indicates that the model detected a high proportion of the relevant objects.
- mAP@0.5 of 0.9503 shows strong detection performance at IoU 0.50.
- mAP@0.5:0.95 of 0.667 is stricter and highlights remaining localization challenges.

Evaluation outputs include the confusion matrix, normalized confusion matrix, training curves, precision/recall/F1 curves and PR curve.

9. Inference on Unseen Data

The final model was tested on unseen test images to check how it behaved outside the training examples. The saved prediction results contain bounding boxes, class labels and confidence scores.

Inference Item	Result
Model used	models/best.pt
Input	Unseen test images
Saved prediction images	20
Confidence threshold used for demo	0.35
Video inference	Performed

The project includes more than the minimum 15 prediction images requested by the internship task. Video inference was also performed as an additional demonstration feature.

10. Error Analysis

Manual review of the unseen test predictions identified ten incorrect or weak cases. The table below uses the requested format and records the actual object, model prediction, error type and likely reason.

Image	Actual Object	Model Prediction	Error Type	Possible Reason
Image 11	Vests of two workers	Vests not detected	False negative	Workers are relatively distant/small in the image.
Image 62	Vests visible in reflection	Vests not detected	False negative	Reflection makes the objects less clear and harder to localize.
Image 65	Helmet visible in reflection; partially visible helmet behind bushes	Helmets not detected	False negative / small-object issue	Reflection, partial visibility, and occlusion make the helmets difficult to detect.
Image 69	Helmets of distant workers	Helmets not detected	False negative	Small and distant objects provide limited visual detail.
Image 74	One person without a vest	Two overlapping boxes for the same person	Poor bounding boxes	Overlapping regions and ambiguous object boundaries caused duplicate detections.
Image 85	Helmets visible in reflection	Helmets not detected	False negative	Reflection changes the visual appearance of the helmets.
Image 88	Blurred person's helmet and separate blurred person	Helmet missed; blurred person grouped into another bounding box	False negative / poor bounding box	Blur and occlusion make separate objects difficult to distinguish.
Image 5	Full vest of the middle worker	Only part of the vest was detected	Poor bounding box	The model localized only part of the visible vest.
Image 65	Tree/background; no helmet object	Tree classified as without_helmet	False positive / wrong class	Background appearance was incorrectly associated with the without_helmet class.
Image 62	Distant worker with light falling on the person	Worker not detected	False negative / lighting issue	Strong lighting and distance reduced useful visual features.

Main Error Categories

- False positives
- False negatives
- Wrong class predictions
- Poor bounding boxes
- Small-object detection problems
- Reflection-related problems
- Blur and occlusion problems
- Lighting-related problems

Recommended Improvements

- Add more images for weak classes, particularly without_helmet.
- Improve annotation quality for small, partially visible and overlapping objects.
- Add more low-light and difficult-lighting examples.
- Include more reflection and occlusion examples.
- Train for more epochs when additional suitable data is available.
- Try YOLOv8s instead of YOLOv8n.
- Remove duplicate or highly similar images.
- Balance the dataset across classes.

11. Streamlit Demonstration

A Streamlit application was prepared as the project demonstration. It loads the final YOLOv8 model and provides image and video inference. A confidence-threshold control is included so that the user can adjust how selective the displayed detections are.

Demo File	Purpose
1det.JPG	Detected image example
2det.JPG	Detected image example
3det.JPG	Detected image example
demooooo.mp4	Video inference example

The demo evidence is stored in `outputs/demo_results/`, while the application source is stored as `app.py` in the project root.

12. Final Project Structure and Deliverables

```
Helmet_Safety_Detection/  
■■■ dataset/  
■■■ notebooks/  
■ ■■■ training_notebook.ipynb  
■■■ outputs/  
■ ■■■ training_results/  
■ ■■■ predictions/  
■ ■■■ demo_results/  
■■■ models/  
■ ■■■ best.pt  
■■■ report/  
■ ■■■ final_report.pdf  
■■■ app.py  
■■■ README.md  
■■■ requirements.txt
```

Required/Prepared Item	Status
Dataset + YOLO labels	Completed
data.yaml	Completed
Training notebook	Completed
YOLOv8n best.pt	Completed
Training artifacts	Completed
Evaluation graphs/matrices	Completed
20 unseen prediction images	Completed
10-case error analysis	Completed
Streamlit image/video demo	Completed
README.md	Completed
requirements.txt	Completed
Final report	Completed

13. What I Learned

- How to prepare and organize a Computer Vision object-detection dataset.
- How to review and standardize YOLO-format annotations.
- How YOLO class IDs and normalized bounding boxes work.
- How to train YOLOv8 using pretrained weights in Google Colab.
- How to evaluate object detection using precision, recall and mAP.
- How to interpret confusion matrices and training curves.
- How to perform manual error analysis and identify model limitations.
- How to package a trained model into a simple inference demo.

14. Future Improvements

- Increase dataset diversity and add more difficult examples.
- Improve annotation quality and consistency.
- Add more low-light, reflection and small-object examples.
- Increase representation of weak classes such as without_helmet.
- Experiment with YOLOv8s or another larger detector.
- Tune inference thresholds using validation evidence.
- Test the system on real-time CCTV/video streams.
- Continue reviewing false positives and false negatives after each model update.

15. Conclusion

This project demonstrates an end-to-end Computer Vision object-detection workflow using YOLOv8n. The final system detects four safety-related classes and includes dataset preparation, annotation review, model training, quantitative evaluation, unseen-data inference, manual error analysis, a Streamlit demonstration and final project documentation.

The recorded evaluation results were Precision 0.9517, Recall 0.9016, mAP@0.5 0.9503 and mAP@0.5:0.95 0.667. The error analysis shows that the main remaining challenges are small/distant objects, reflections, blur, occlusion, lighting variation and ambiguous object boundaries. These findings directly inform the proposed future improvements.

The report intentionally does not include a LinkedIn/portfolio section. It is focused on the internship task requirements and the actual work completed in the Helmet Safety Detection project.